

Title: The benefits and barriers of AI for Persons with Disabilities: A scoping review
Author(s): Diana Drozdzik, Steven Little and Yash Chopra
Affiliations: SFU and GLOCAL
Date of submission:

Abstract

Advancements in AI-driven assistive technology offer unprecedented opportunities for improving the lives of persons with disabilities. This scoping review examines the benefits, barriers, and recommendations for AI technologies in addressing the challenges faced by disabled individuals. The findings highlight AI's potential in diagnostic improvement, assistive chatbots, adaptive communication tools, and workplace integration. However, risks such as bias in AI models, privacy concerns, and lack of inclusive design present challenges to ethical and equitable AI adoption. Recommendations for user-centered design, governmental oversight, and adoption of the social model of disability are indicated as measures to help ensure that AI-driven technologies enhance, rather than hinder, accessibility and inclusion for those with disabilities.

Introduction

Understanding disability literature is imposed upon by the variable definitions of disability that are across countries and regions. In British Columbia, Canada, having a disability is defined by obtaining a government status called Persons-with-Disability (PWD). To be eligible for this status, one must be older than 18, have a severe physical or mental impairment that is expected to continue for more than 2 years, be significantly restricted in one's ability to perform daily tasks, require assistance with daily living activities from another person, an animal or assistive device, and require financial assistance (Ministry of Social Development and Poverty Reduction, 2025). Disabilities can be physical (mobility, sensory), intellectual (cognitive, developmental), mental health-related (anxiety, depression), or invisible (chronic pain, neurological conditions), each affecting daily life in different ways.

More than 1 out of 4 (27%) of Canadian citizens over the age of 15 have a disability, an increase in percentage from 22% since 2022 (Government of Canada, Statistics Canada, 2024). The likelihood of acquiring a disability increases with age, jumping from a 20.1% prevalence rate for those under 24 years to 40.4% after the age of 65 (Statistics Canada, 2023). In Canada, disabilities are correlated with higher poverty rates, reduced workforce participation (Statistics Canada, 2024), less social contacts, and lower satisfaction with social contacts than those without disabilities (Statistics Canada, 2023). These factors can affect the mental health and overall well-being of individuals. Despite many resources to aid the challenges that disabled individuals face, surveys reveal that barriers to accessibility, difficulty accessing services, and unmet medical needs still remain significant in the demographic (Statistics Canada, 2024).

When considering solutions to disability accessibility, it is critical to be mindful of the type of lens that is being used to look at the problem. For example, focus on physical abnormalities of disabilities prioritises "fixing" the abnormality rather than focusing on solutions that work

with the disability as it is. This approach can place undue weight on the individual and emphasise their difference, instead of shifting focus to how community infrastructure can be built inclusively. Crip Spacetime, a theory developed by analysing data from the Disabled Academics Study which included over 250 disabled respondents and in-depth interviews with thirty-eight disabled academics, emphasises how the temporality (time-related) and spatial (space-related) dimensions of life are shaped by an individual's disability, particularly highlighting experiences that deviate from able-bodied norms. This framework moves away from focusing solely on individual disabled bodies and the obsession with accommodating them, instead focusing on relations, systems, objects, and discourses. Crip Spacetime offers a lens for a more inclusive understanding of the lived experiences of people with disabilities and for involving these experiences in the solutions creation process (Price, 2024).

This shift in focus can be understood through the lens of upstream versus downstream effects. Downstream effects are the issues that manifest at an individual level, such as a disabled student requiring extra time for an exam or a wheelchair user needing a ramp to enter a building. Accommodations often address these downstream effects, providing individualised solutions to immediate access barriers. However, upstream effects are the systemic and societal factors that create these barriers in the first place, such as ableist assumptions in curriculum design or a lack of universal design principles in architecture.

While accommodations are often appreciated and necessary in the present, they can be viewed as "Band-Aids" that treat the downstream symptoms of a fundamentally inaccessible system. Crip Spacetime advocates for addressing the upstream causes of inequity by critically examining and dismantling the "violent operations enabled through ideologies of ability" (Price, 2024). Crip Spacetime can offer insights into how technologies, such as AI, might impact the temporal and spatial access, barriers, and opportunities for people with disabilities, though it's crucial that these do not merely become further downstream solutions without addressing the upstream systemic issues (Price, 2024).

With this theoretical framework in mind, there is an opportunity to explore new possible solutions to helping PWD find greater accessibility and improved medical satisfaction. With the recent boom in artificial intelligence (AI) across private and public sectors, there is a spectrum of new technologies that warrant investigation into their potential in achieving this. AI refers to computer systems capable of performing tasks that typically require human intelligence. These tasks include learning, reasoning, problem-solving, perception, decision-making, and understanding natural language. AI systems are designed to adapt, improve, and operate autonomously or semi-autonomously.

While AI holds great potential across many industries, it is important to consider the environmental impacts it can have. Generative AI hardware is known to merit substantial impacts, specifically in the manufacturing of graphics processing units (GPUs) required to run GenAI system. Notably, GPUs are required to train the sophisticated algorithms that Large language models (LLMs) use, such as with ChatGPT (Valdivia, 2024). The lifespan of GPUs is usually 5 years (European Commission, Joint Research Centre, 2020), and resources needed to produce these units include: water, electricity, rare metals like tantalum and cobalt (European Commission, Joint Research Centre, 2020). Computational hardware produces significant heat and requires large amounts of water to cool the servers. GPUs and CPUs also contribute to e-waste because they are non-recyclable (Valdivia, 2024).

Environmental impacts aside, AI is becoming more widespread. The expectations and demands of citizens who come into contact with it are spreading across private and public sectors, creating market pressure (Madan & Ashok, 2024). It follows that the large demographic that constitutes disabled individuals can come to expect the personalization and efficiency of AI to spread into more areas of and functions their lives. To help inform this process, we see a need to review the current state of AI in how it interfaces with persons with disabilities, as well as how the future of AI can be shaped to better serve this population.

Research Questions

- 1) What AI technology is available or being developed that can benefit persons with disability and the unique struggles they face?
- 2) What challenges, barriers, and risks exist for persons with disabilities in the context of AI technology?
- 3) How can AI technology be improved to reduce risks, mitigate barriers, and better serve persons with disabilities?

Methodology

Literature Review

For our scoping review we used a variety of databases to conduct research in order to understand how AI intersects with disabilities.

To guide our review and inclusion criteria, we followed the PICO (population, intervention, comparison and outcome) process. A reliability check was done by reviewing each other's search keywords and acquired papers. We used the SRQR and PRISMA checklist to improve the quality of our qualitative research.

Eligibility Criteria

In this integrative review we focused on the population of individuals identifying as disabled. We conducted our review with the intention of covering many types of AI interventions for comparison. Our comparison groups are alternative forms of AI interventions and our outcomes are: health outcomes, accessibility to resources, and social inclusion. To ensure our review is up to date with contemporary technological advances, we excluded papers that were older than five years. As well, to improve accuracy and reduce the chance of replicating bias, we limited the use of other reviews to draw data for our literature review.

Search Strategy

Study databases that were included in our search were:

Medline (OVID), Medline (PubMed), CINAHL, Sage, SFU's library catalogue, and Google Scholar to help widen our search.

Keywords were taken from articles but mainly were:

disability/disabilities/PWD/person with disabilities/ person-with-disability,
AI/artificial intelligence/chatbots/conversational agent/accessibility/machine learning

Information Source

The primary databases were Medline (OVID) and Google Scholar. Secondary databases were CINAHL, Medline (PubMed), and Sage. Papers included stakeholder interviews, surveys, randomized controlled trials, semi-structured interviews, observational studies and some scoping reviews for broader understanding.

Data Collection

Patient demographics included a wide variety of disabilities. We did not place a limitation on types of disability. Age groups ranged from 15 to beyond 65 years old(refer to Figure 1.1)). On average there were slightly more women in the patient groups. The most dominant age group were individuals 65 years and older. We had a variety of disability types represented in our review, with seeing-related disabilities as the most dominant disability type. The standard deviation in representation across disability types was quite small, with the deviations from the mean range from +2.2 to -1.8 percentage points (refer to Figure 1.2). All types fall roughly between 17-21%. Disability types included are: sight-related, dexterity, flexibility, mobility, developmental, hearing, learning, memory and mental health-related. Canadian studies were favoured for healthcare system relevance to this study's country of origin. We used a purposive sampling method, where papers were intentionally chosen based on desired characteristics.

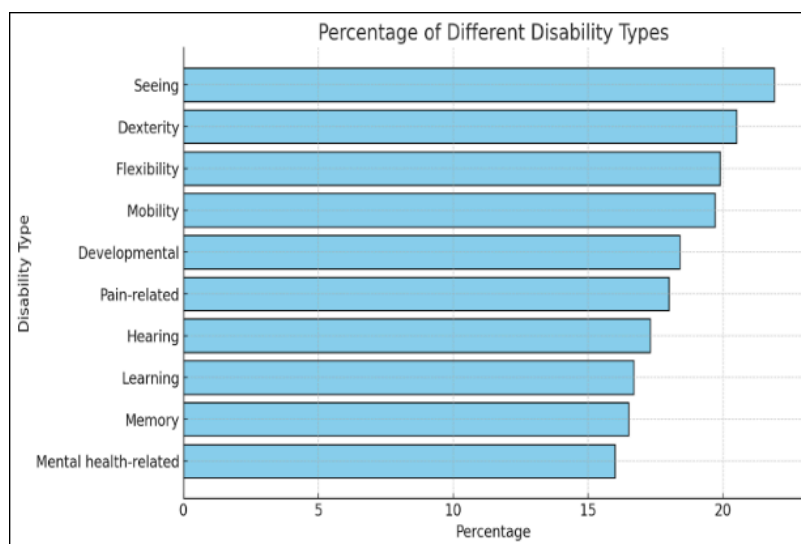


Figure 1.1: A graph indicating the percentage of different disability types represented in Canada. Note that individuals can have multiple types of disabilities overlapping. (Statistics Canada)

Strengths

Diverse Study Designs

Studies included in the review are diverse in study designs, such as randomized controlled trials, observational studies and case reports. Each study design provides unique insights and may strengthen findings.

Limitations

Missing Context

Data does not show:

- Comorbidities between disability types
- Severity levels within disability categories
- Age of onset information
- Whether the disabilities are primary or secondary

Aggregation Challenges

When synthesizing findings, different studies might have defined and categorized disabilities differently depending on the country or context.

Category Overlap

The disability types, such as 'pain-related', 'mobility' and 'flexibility' likely have significant overlap, which can make it difficult to make clean comparisons between the different types.

Areas for Improvement

More studies that are randomized control studies would offset the number of reviews used and add more depth in the results and discussion section. Finding more studies that are surveys, focus groups or interviews can provide more data for our outcomes: accessibility, health outcomes, and social inclusion.

Gender Binary Representation

Studies only show men and women, which is potentially missing data for those who do not identify as male or female, such as non-binary, transgender or two spirited.

Missing Youth Data

The absence of individuals under 15 can be a limitation when considering comprehensiveness. However, individuals under the age of 15 are not normally independent regardless of whether they have a physical status synonymous with being disabled.

Researcher Characteristics and Reflexivity

This paper was created by three bachelor graduated individuals with mentorship from GLOCAL. Our team includes individuals who have had a disability or are currently considered disabled. One individual in our team has PWD status and will not be disclosed for their privacy. Disabilities present in this research group are considered invisible disabilities which is a different lived experience than someone with a mobility or developmental disability. Our study design was affected by an inability to receive REB qualifications for submitting our research, so we chose a design that did not need REB approval. To align with our intention to provide evidence-based change to Canada's disability system and policies, we chose more Canadian studies in our review. We received peer debriefing from another researcher to mitigate potential bias.

Context

This research is primarily situated in Canada, specifically British Columbia during the AI revolution boom across private and public sectors. There is a market push for AI adoption

across industries for a spectrum of user demographics. AI is a relatively young field and should be approached cautiously. The rationale for this study stems from a need for better accessibility and health satisfaction for those with disabilities.

Techniques to enhance trustworthiness

Member checking of data analysis and a chart with all literature review papers with notes were used to ensure data quality.

Results

Overview

This section presents the findings from our scoping review, which sought to explore the intersection of disability and artificial intelligence (AI), with a focus on both barriers and opportunities for AI applications in improving the lives of persons with disabilities. Based on our research questions, we focused on three overarching themes that emerged from our review: 1) Benefits & Potential, 2) Challenges, Risks, & Barriers, and 3) Recommendations for Inclusive AI Design.

Benefits and Potential

Diagnostic Improvement

Across many developments, artificial intelligence shows great promise in its potential to improve the lives of persons with disabilities. AI has many applications in the realm of diagnosis. AI models can extract and process patient data, using state-of-the-art machine learning algorithms such as decision trees, support vector machines, and deep learning to expedite and improve the accuracy of diagnosis. Through its advanced pattern-finding abilities, AI is able to integrate many data sources at once and even discover new correlations. For example, natural language processing (NLP), a technique used to process human language, can be used to identify speech patterns that may indicate the presence of an intellectual disability (Almufareh et al., 2023). El Morr et al. (2024) found that AI has been documented in the accurate diagnosis of multiple sclerosis (MS), developmental disorders, and autism, all of which can be significantly disabling. Prognosis and prediction of disability also show to be promising use cases of AI technology, having shown to predict the risk of reading disabilities among students (Erbeli et al., 2023) and the likelihood that injuries become chronic (Koc et al., 2021).

Chatbot-based Treatments

Beyond diagnosis, AI has many applications in assistive technologies to help persons with disabilities manage symptoms, find greater accessibility, and improve their quality of life. Conversational agents, otherwise known as chatbots, are one of the most well-known AI technologies and show great promise for a number of disabilities. Chatbots are easily accessible and cost-effective, and can provide a variety of services to disabling users (Guo et al., 2019). In the domain of psychiatric disorders, chatbots providing cognitive-behavioural

therapy have been shown to improve user-reported measures of psychological well-being, anxiety, depression, or perceived stress compared to controls across a range of disorders. As well, they show an advantage in increasing adherence to therapy (Guo et al., 2019). Jang et al (2021) found that a chatbot built for ADHD management to be more effective at improving measures of hyperactivity-impulsivity and other ADHD symptoms than a textbook offering the same techniques. Deafness, another condition that can be disabling, may benefit from the use of chatbot sign language through integrated video capabilities, or through offering real-time captions of live audio (Almufareh et al., 2023). As well, text-to-speech and Automatic Speech Recognition (ASR) capabilities can be aids to users with visual impairments (Guo et al., 2019), and are accessible features for chatbot integration. Mateos-Sanchez et al. (2022) found benefit to users with intellectual disabilities to improve communication and social skills through chatbot-integrated training. Looking forward, chatbots can be projected to provide cognitive assistance to those with intellectual disabilities, dementia, or other disorders that affect cognitive function (Guo et al., 2019), or integrate facial recognition technology with therapies to help identify emotions in disabled users (Almufareh et al., 2023).

Robotics, Wearables, and Virtual Assistants

Outside of empowering diagnosis and treatment, there is a range of encouraging developments in technology that leverages AI to assist disabled individuals. AI can be integrated into wheelchairs and other mobility devices to improve navigation and control features (Olawade et al., 2025), or be integrated into robotics to help disabled people with everyday tasks, provide support, and offer companionship (Almufareh et al., 2023). Virtual assistants with AI advancements can help reduce access barriers to everyday tasks like telecommunications, transmitting messages to caregivers, and coordinating grocery deliveries. Developments in virtual reality and wearable devices may offer exciting new support for rehabilitation and exercise in disabled individuals (Olawade et al., 2025).

Communication

In disabilities that affect communication and language, technology that leverages AI has a range of applications. For deaf individuals, AI-enhanced real-time captioning can generate live captions to translate audio, and for those with an intellectual disability that affects language abilities, AAC (augmentative and alternative communication) devices can aid with verbal communication through text-to-speech capabilities and visual representations of phrases, words, and concepts (Almufareh et al., 2023). As well, AI systems built for facial recognition can help the intellectually disabled identify, interpret, understand and communicate their feelings in ways they may otherwise struggle with. In disabilities that impact reading and writing such as dyslexia or dysgraphia, text analysis systems can aid them through visual illustration, focused highlighting, intelligent spelling, grammar correction, and word or phrase suggestions (Almufareh et al., 2023).

Education & Workplace

AI possesses the ability to create custom-tailored, adaptable, and personalized learning plans for those with learning disabilities (Almufareh et al., 2023). This AI-powered educational tool has been shown to increase engagement and motivation to learn in students when compared with conventional learning approaches (Mateos-Sanchez et al., 2022). In the

workplace, AI tools have potential to create more inclusivity and accessibility by reducing barriers, offering personalized solutions, and lowering accommodations costs for employers (Disability Inclusion Business Council, 2024).

Social Benefit

Across its range of benefits, AI technology has the power to help disabled individuals overcome barriers and find greater health equity and inclusion in society. AI technology could facilitate participation in community through training social skills (Mateos-Sanchez et al., 2022), allow accessibility to information when used for disabilities that affect communication and hearing (Apuzzo & Burrese, 2022), and create workplaces that are more inclusive and accessible through personalized assistive technologies (Disability Inclusion Business Council, 2024). AI-based at-home therapy can combat cost-based service inaccessibility (Almufareh et al., 2023), and through improving accessibility, reduce social disparities that result from technology being otherwise unusable to certain disabled populations (Guo et al., 2019).

Challenges, Risks, and Barriers

Although offering hope to benefit those with disabilities, artificial Intelligence is still a relatively young field. Without deliberate attention to the design of technologies that integrate it, AI runs the risk of perpetuating bias, discrimination, and disparities as well potential privacy concerns.

Inaccessibility of Technology

AI technology available to the general public may leave behind users of different disabilities if it does not accommodate their uniqueness. For example, AI systems that use natural language processing (NLP) to understand text inputs from users may provide less accurate results for those with cognitive or intellectual disabilities, and automated speech recognition systems may function poorly for those with speech impairments. Those with conditions that affect their facial features, body morphology, or motor movements may run into issues with systems that use this data to function, such as with gesture recognition systems and facial analysis software. Furthermore, photo analysis AI may not be accessible to those with tremors or visual impairment who struggle to take quality photos that the system is designed for (Guo et al., 2019).

Though chatbots show promise to help persons with disabilities, those with cognitive, communication, and coordination barriers may face challenges that limit their ability to navigate them. The World Health Organization's W3C Web Accessibility Initiative highlights cognitive disability barriers to using technology: memory, executive functions, reasoning, attention, language, knowledge and behavior.. These barriers can be related to cognitive ability domains: 1) language, communication, and auditory reception 2) reasoning, ideation, and cognitive speed 3) memory and the ability to learn 4) visual perception and 5) knowledge and achievement. These categories can show a wide array of areas that can limit individuals with a cognitive disability in accessing healthcare and using tools such as AI chatbots to help

increase access to health resources. Ultimately these categories can impact an individual's literacy level when interacting with AI, AI chatbots and chatbots (Tielman et al., 2024).

Biases in AI

In their development, AI models are trained on data to help them understand, process, and generate content. If this training data lacks the diversity that is exemplified by persons with disabilities, AI models can run the risk of perpetuating bias against the demographic (Almufareh et al., 2023, El Morr et al., 2024, Guo et al., 2019, Disability Inclusion Business Council, 2024). AI systems that are trained on aggregated training metrics that do not adequately represent persons with disabilities can lead to inaccurate responses or generalizations (Guo et al., 2019). Due in part to mechanisms like this, Newman-Griffis et al. (2023) show how different conceptualizations of disability can lead to highly distinct AI technologies with divergent biases, ideologies, and implications. Urbina et al. (2025) demonstrated examples of how popular AI large language models (LLMs) ChatGPT and Gemini give responses that underestimate the prevalence of disability in the general population, and were more likely to describe those with disabilities in unfavorable language. These findings suggest bias in the training data and design of these systems, and run the risk of perpetuating inequalities in society and medical settings. Unrepresentative AI design and training data can also discriminate persons with disabilities by erroneously detecting them as anomalies, as in the case of AI bot-detection systems labelling disabled users as outliers due to their inability to solve puzzles like CAPTCHAs with a response time that the algorithm perceives as "human" (Guo et al., 2019) (Olawade et al., 2025).

Bias and discrimination from AI is particularly consequential in how it can manifest in the workplace. The increasingly popular trend of online hiring and recruitment algorithms may mean that by default disclude certain disabilities if they lack accessibility options. These algorithms may also select for a narrow criteria that disabled people are less likely to fall under, neglecting other qualifying factors. As well, they may select against writing styles that may vary across disabilities, potentially even detecting their job applications as spam (Olawade et al., 2025). Moreover, as jobs become more automated due to advancements in AI, employees with disability may be disproportionately negatively impacted due to their overrepresentation in jobs that are more easily automated (Disability Inclusion Business Council, 2024).

Data Privacy

AI's reliance on user data to learn and improve poses a particular risk to disabled users, who may share sensitive personal health information (Olawade et al., 2025) and who may be more easily identifiable in datasets due to having unique identifiers associated with the barriers they face in areas like work experience and education (Disability Inclusion Business Council, 2024). If compromised, their sensitive data may then be at risk of the consequences of data breaches or misuse by companies (Olawade et al., 2025). Because of the vulnerabilities that PWD face with their data, it is crucial that data privacy and security is prioritized in the design and development of AI systems (Almufareh et al., 2023).

Recommendations for Inclusive AI Design

The Social Model of Disability

At the root of disability is how it is defined, and in turn, how its definition can shape action and policy. The Convention on the Rights of Persons with Disabilities (CRPD) defines disability in accordance with the social model of disability, which understands disability as an interaction effect between individuals and their environment, and how this interaction can either obstruct or empower their involvement and participation in society. However, within AI research there is a lack of use of this social model of disability (El Morr et al., 2024). On the other hand, the medical model of disability views impairment as an individual struggle that needs to be fixed for the individual to fit societal bodily norms. For an individual with a lung disability, the social model of disability would look at changing the environment to accommodate this difference, whereas the medical model of disability focuses on changing the individual, such as providing a lung transplant (Berghs et al., 2016). Another example can be found in how nearsightedness and farsightedness are so normalized and accommodated that they are not considered disabilities, displaying the social model of disability's perspective of how disability is socially constructed. The adoption of the social model of disability can help set a framework in the design phase of AI technology which shifts the burden away from disability as a medical diagnosis, instead being an aspect of human diversity that inclusive design can accommodate and support (Olawade et al., 2025).

Governmental Oversight

Encouraging governments to uphold the human rights of those with disabilities is another way that the future of AI can promote inclusivity, reduce bias, and improve accessibility, rather than diminish it. An example of this is Digital Nations, an international forum that includes Canada amongst 10 countries with the commitment of steering new technologies in a way that improves citizens' lives (Treasury Board of Canada Secretariat, 2024). Among its core principles are the fostering of digital inclusion and accessibility, which help inform Canada's policies around AI development such in its Directive on Automated Decision-Making and in Digital Charter Implementation Act. Within the rapidly developing realm of AI technology, it is crucial that governmental entities across the world work to discuss, regulate, and enforce design principles that can help accommodate and support those with disabilities (Disability Inclusion Business Council, 2024). Almufareh et al. (2023) recommends that disability advocacy groups advocate for the creation and enforcement of government policies and regulations to help ensure that AI upholds the human rights and social equity of those with disabilities.

To protect the personal information of persons with disabilities, there is a need for strong encryption protocols to protect the data of those with disabilities both if it is used for AI training or if used during AI technology use (Lanzi et al., 2023). Those with disabilities may be more vulnerable to data-related exploitation (Olawade et al., 2025), and it is essential to protect their data through compliance with national regulations such as HIPAA in the United States, or Europe's General Data Protection Regulation (GDPR) (Guo et al., 2019).

Inclusive & Ethical Design

On the corporate and business level, promoting stakeholder interest in the creation and implementation of inclusive and fair AI systems can help to offset potential biases that

impact disabled individuals. Executives, human resource workers, and IT professionals can consider whether an AI system used in the workplace matches organizational values, evaluate its outcomes, and investigate its training data (Olawade et al., 2025).

To support ethical design, algorithmic bias (such as AI models being trained off data that does not sufficiently include disability and the diversity it represents) can be addressed. Apuzzo & Burresi (2022) proposes that LLM models should be trained on reweighted datasets to deliberately consider sensitive demographics that may be affected by bias. Similarly, to help ensure that AI models are more balanced and representative, Almufareh et al. (2023) suggest artificially inflating the number of samples for commonly underrepresented classes of people.

User-centered design is a core principle that can help systems align with the diversity of disability and promote inclusivity. User-centered design puts users at the center of product design and development. In the context of AI and disability, this involves including disabled individuals from the early stages of research and design, iteratively incorporating their feedback, and testing the system with the demographic (Lanzi et al., 2023). One way to support this process is to have interdisciplinary collaboration between AI researchers, disability experts, and disabled individuals throughout AI development (Nugent et al., 2022; Olawade et al., 2025). By focusing on user-centered design, AI systems may be able to mitigate bias, improve accessibility, and reduce potential discrimination for disabled people.

In the design of chatbots or self-management programs, it is essential to address the various barriers of disabilities by tailoring tools to different disability types. This means creating a feature to identify the type of disability of the user and adapt their output to align with their needs. Creating explainable artificial intelligence (XAI) systems that transparently convey information to the user about what it is doing, how it is doing it, and why, can significantly improve inclusivity of users. XAI must also accommodate deaf and blind populations by incorporating accessible mediums for effective communication (Tielman et al., 2024).

Discussion

From our review, we identified 3 major areas that AI technology can help to improve the lives of persons with disabilities: accessibility, health outcomes, and social inclusion.

Accessibility

Canada's Disability Action Plan defines accessibility as "the degree to which a product, service, program or environment is available to be accessed or used by all" (Government of Canada, 2022). Those with disabilities can have unique challenges with day-to-day accessibility, instead running into a spectrum of barriers that hinder their full and equal participation in society. Addressing accessibility involves considering the context of the resource, identifying barriers to its use, and understanding that accessibility improves as these barriers are mitigated. *The Accessible Canada Act* (Government of Canada, 2024) lists 7 major priority areas for improving accessibility for Canadians:

1. employment
2. the built environment

3. information and communication technologies
4. communication (other than information and communication technologies)
5. the procurement of goods, services and facilities
6. the design and delivery of programs and services
7. transportation

Our literature review reveals a wide array of AI technology and directions for future AI development which may be able to help address key areas of Canada's accessibility priorities:

Employment

Employment opportunities for PWD in British Columbia are constrained by a lack of inclusive workplace cultures, further limiting financial independence (Disability Inclusion Business Council, 2024). Data from Statistics Canada (2017) highlight additional barriers, including lack of transportation, limited educational opportunities, and inadequate workplace accommodations.

Adopting inclusive design principles is critical to creating workplaces that exceed accessibility standards (Disability Inclusion Business Council, 2024). Recommendations include fostering leaders who champion accessibility and implementing governance plans to ensure fair use of technology in hiring.

AI could help create inclusive workplaces through personalized assistive technologies by providing real-time communication support and adaptive interface customization. These technologies can enable independent task completion, streamline workflow management, and facilitate more effective collaboration between team members.

Information and Communication Technologies

AI shows strong potential in its ability to help persons with disabilities engage seamlessly with information and communication technologies. By encouraging both policy and stakeholder interest to support the design of technologies that are inclusive, bias-controlling, and non-discriminatory, these technologies can help serve a wide spectrum of disabilities across populations. Adopting user-centered design and implementing AI capabilities can allow systems to adapt their interface and features to accommodate a range of impairments and personal preferences. For example, a business creating a mobile application for ordering groceries may expand their market and user-base by implementing user-centered options that allow for various engagement needs. The application may include options for the visually impaired with speech-to-text and text-to-speech capabilities, gesture or expression-based interaction for those with motor impairments, or an interactive chatbot that can assist with the process of ordering groceries for those with cognitive disabilities.

Communication (Other than ICT)

AI can address numerous communication barriers faced by those with disabilities. For the deaf and hearing impaired, systems that can understand and translate this into speech could open up the potential for reduced communication barriers with those who do not understand sign language. Those with disabilities that affect their emotional processing may better be able to understand and convey their emotional state to others with the use of AI that can

interpret facial expression or analyze their text for pertinent patterns. Or, if an intellectual disability makes verbal communication a challenge for an individual, an augmentative and alternative communication device may help infer their desired message with the help of symbol-based and text-to-speech capabilities.

Procurement of Goods, Services and Facilities

The resources required by persons with disabilities vary by subgroup but generally include healthcare, specialized services, and rehabilitation. These resources improve health outcomes and promote independence (Lanzi et al., 2023). Financial independence, supported by employment or stable income, plays a critical role in achieving autonomy (Disability Inclusion Business Council, 2024).

Experts emphasize the importance of well-being self-management programs—both online and in-person—that are accessible and designed to transition persons with disabilities from dependence to self-sufficiency (Lanzi et al., 2023). AI offers a new and exciting potential for effective at-home management of disabilities. Chatbot-based therapies help lower the barriers of cost and transportation and can empower disabled users to rehabilitate through technologies that adapt to their needs.

Through AI helping to improve technological accessibility, disabled users can become more independent to use web and mobile-based options for connecting them to services like health appointments, transportation, and grocery delivery. Technology like virtual assistants can further assist those with disabilities in connecting to the goods and services they need.

Design and Delivery of Programs and Services

To improve the design of programs and services, consultation with disability-familiar interdisciplinary teams and the implementation of user-centered design is essential. Innovative learning assistance using adaptive and personalized approaches can help shape how education is delivered for disabled individuals. As well, at home health management systems that utilize chatbots and integrative AI features have the potential to make the delivery of healthcare more widely available and accessible to a range of disabilities.

Health Outcomes

Health outcomes, including improvements in physical, mental, and emotional health, reflect the direct impact of AI technologies on well-being. AI-based tools such as conversational agents and personalized health management systems have shown measurable benefits. For example, Jang et al. (2021) highlighted significant reductions in ADHD symptoms, including impulsivity and hyperactivity, among users of an AI-driven psychoeducation chatbot.

AI-powered assistive devices have also been effective in rehabilitation and therapy. Almufareh et al. (2023) reported on AI tools designed to support cognitive assessments and personalized therapy, enabling better patient outcomes through tailored interventions. Furthermore, Olawade et al. (2025) outlined the potential of AI-enabled robotics and multimodal systems in providing personalized care and improving functional independence among people with disabilities.

Social Inclusion

According to the social model of disability, disability is defined by how an individual is either supported or hindered by society. The level to which a disabled individual is accommodated or integrated links to their sense of social inclusion and belonging. Statistics Canada (2023) found that those with mental health disabilities have less social connection and satisfaction than those without such disabilities. This highlights a need to support social well-being and inclusion for this demographic.

Efforts to foster disability-inclusive communities have included virtual platforms, such as Wiki-disability, which provides data for international and national disability advocacy committees (El Morr et al., 2021). While these platforms could be valuable help to disseminate knowledge about disability rights, they do not necessarily create social spaces that reflect the lived experience of being disabled.

According to the Bridging the Gap Report (Disability Inclusion Business Council, 2024), AI solutions and digital tools have demonstrated significant potential in creating more inclusive and accessible environments across various aspects of daily life. Our findings suggest that AI has significant potential to enhance the participation of individuals with intellectual disabilities who have historically faced barriers to community participation due to challenges with social skills development, chatbots offer innovative educational resources that facilitate real-world integration. These tools provide structured opportunities to learn and practice social skills through active methodologies, helping bridge the gap between educational settings and practical community engagement.

In the workplace, AI-powered assistive technologies can address multiple accessibility barriers simultaneously. Accessible and personalized accommodations may be able to make integration into the workplace more feasible and sustainable for those with disabilities. Our findings indicate the deaf community as a case for which AI could offer a key role in aiding social integration, with tools like specialized communication assistance.

However, it is key to note these approaches align more with the medical model of disability, which focuses on treating or lessening the severity of a disability rather than addressing the societal structures that create barriers in the first place. Social accommodations, unlike AI-driven solutions, remain complex due to the sheer diversity of human experiences and environments. The vast span of use-cases for AI is encouraging, but it is important that our societal perceptions of disability encompass its complexity so that these emerging technologies can effectively and ethically serve the disabled community.

Conclusion

This systematic scoping review revealed a large array of benefits, challenges, risks, and future indications pertaining to artificial intelligence and persons with disability. AI-powered technologies including chatbots, assistive communication tools, and diagnostic systems offer promise in improving healthcare and accessibility for disabilities. However, the risks of algorithmic bias, data privacy vulnerabilities, and exclusionary design must be addressed to ensure equitable AI development. Because of both the large potential positive and negative impacts it can have, there is an apparent need for policy-makers, stakeholders, developers, experts and disabled individuals alike to cross-collaborate in achieving accurate, accessible, inclusive, and user-centered AI technologies. Governments should continue to develop evidence-based and comprehensive policy frameworks to guide ethical AI implementation. If AI is implemented with careful consideration of bias-filtering practices, ethical implications, and inclusive design principles, it can capitalize optimally on improving daily accessibility, health outcomes, and social inclusion for persons with disabilities. Without caution, however, AI runs the risk of perpetuating social inequity and discrimination towards this already marginalized population.

Funding

The research described in this paper has been supported by mentorship and funding from GLOCAL Foundation of Canada in their "CANConnect: Micro-grants Diversity for Canadian Youth" program.

Conflicts of Interest:

No conflicts of interest were present.

Bibliography

Almufareh, M. F., Tehsin, S., Humayun, M., & Kausar, S. (2023). Intellectual disability and technology: An artificial intelligence perspective and framework. *Journal of Disability Research*, 2(4). <https://doi.org/10.57197/JDR-2023-0055>

Apuzzo, C., & Burresi, G. (2022). Designing accessible chatbots for deaf people. *2022 11th Mediterranean Conference on Embedded Computing (MECO)*, 1-4. <https://doi.org/10.1109/MECO55406.2022.9797093>

Berghs, M., Atkin, K., Graham, H., Hatton, C., & Thomas, C. (2016). Implications for public health research of models and theories of disability: A scoping study and evidence synthesis. *Public Health Research*, 4(8). <https://doi.org/10.3310/phro4080>

Choi, R. (2021). *Accessibility findings from the Canadian Survey on Disability, 2017*. Statistics Canada.

Deerinwater, J., Ho, S., Thompson, V., Wong, A., Erevelles, N., Morrow, M., Erevelles, N., Morrow, M., Buettgen, A., Zubrow, E., Rioux, M. H., & Viera, J. (2024). A Conversation on Disability Justice and Intersectionality. In *Handbook of Disability* (pp. 503–524). Springer Nature Singapore. https://doi.org/10.1007/978-981-19-6056-7_95

Disability Inclusion Business Council. (2024). *Bridging the gap: Report on disability inclusion in Canadian workplaces*. Government of Canada.

European Commission, Joint Research Centre. *Critical Raw Materials for Strategic Technologies and Sectors in the EU: A Foresight Study*. Publications Office of the European Union, 2020. https://rmis.jrc.ec.europa.eu/uploads/CRMs_for_Strategic_Technologies_and_Sectors_in_the_EU_2020.pdf

El Morr, C., Kundi, B., Mobeen, F., Taleghani, S., El-Lahib, Y., & Gorman, R. (2024). AI and disability: A systematic scoping review. *Health Informatics Journal*, 30(3), 14604582241285743. <https://doi.org/10.1177/14604582241285743>

El Morr, C., Maret, P., Muhlenbach, F., Dharmalingam, D., Tadesse, R., Creighton, A., Kundi, B., Buettgen, A., Mgwigwi, T., Dinca-Panaitescu, S., Dua, E., & Gorman, R. (2021). A virtual community for disability advocacy: Development of a searchable artificial intelligence-supported platform. *JMIR Formative Research*, 5(11), Article e33335. <https://doi.org/10.2196/33335>

Erbeli, F., He, K., Cheek, C., Rice, M., & Qian, X. (2023). Exploring the Machine Learning Paradigm in Determining Risk for Reading Disability. *Scientific Studies of Reading*, 27(1), 5–20. <https://doi.org/10.1080/10888438.2022.2115914>

Government of Canada. (2024). *Accessible Canada Act (S.C. 2019, c. 10)*. Minister of Justice.

Government of Canada. (2022). *Canada's disability inclusion action plan*. Employment and Social Development Canada.

Guo, A., Kamar, E., Vaughan, J. W., Wallach, H., & Morris, M. R. (2019). Toward fairness in AI for people with disabilities: A research roadmap (arXiv:1907.02227). arXiv. <https://doi.org/10.48550/arXiv.1907.02227>

Jang, S., Kim, J.-J., Kim, S.-J., Hong, J., Kim, S., & Kim, E. (2021). Mobile app-based chatbot to deliver cognitive behavioral therapy and psychoeducation for adults with attention deficit: A development and feasibility/usability study. *International Journal of Medical Informatics*, 150, Article 104440. <https://doi.org/10.1016/j.ijmedinf.2021.104440>

Koc, K., Ekmekcioğlu, Ö., & Gurgun, A. P. (2021). Integrating feature engineering, genetic algorithm and tree-based machine learning methods to predict the post-accident disability status of construction workers. *Automation in Construction*, 131, 103896. <https://doi.org/10.1016/j.autcon.2021.103896>

Lanzi, R. G., Modi, R. A., & Rimmer, J. (2023). A disability-inclusive healthcare-to-well-being translational science framework. *International Journal of Environmental Research and Public Health*, 21(1), Article 18. <https://doi.org/10.3390/ijerph21010018>

Madan, R., & Ashok, M. (2024). Making sense of AI benefits: A mixed-method study in Canadian public administration. *Information Systems Frontiers*. <https://doi.org/10.1007/s10796-024-10475-0>

Mateos-Sanchez, M., Melo, A. C., Blanco, L. S., & García, A. M. F. (2022). Chatbot, as educational and inclusive tool for people with intellectual disabilities. *Sustainability*, 14(3), Article 1520. <https://doi.org/10.3390/su14031520>

Ministry of Social Development and Poverty Reduction. (2025, January 6). Disability assistance. Government of British Columbia. <https://www2.gov.bc.ca/gov/content/family-social-supports/services-for-people-with-disabilities/disability-assistance>

Newman-Griffis, D., Rauchberg, J. S., Alharbi, R., Hickman, L., & Hochheiser, H. (2023). Definition drives design: Disability models and mechanisms of bias in AI technologies. *First Monday*. <https://doi.org/10.5210/fm.v28i1.12903>

Nugent, S. E., & Scott-Parker, S. (2022). Recruitment AI has a disability problem: Anticipating and mitigating unfair automated hiring decisions. [10.31235/osf.io/8sxh7](https://doi.org/10.31235/osf.io/8sxh7)

Olawade, D. B., Bolarinwa, O. A., Adebisi, Y. A., & Shongwe, S. (2025). The role of artificial intelligence in enhancing healthcare for people with disabilities. *Social Science & Medicine*, 364, Article 117560. <https://doi.org/10.1016/j.socscimed.2024.117560>

Pierrès, O., Christen, M., Schmitt-Koopmann, F. M., & Darvishy, A. (2024). Could the use of AI in higher education hinder students with disabilities? A scoping review. *IEEE Access*, 12, 27810-27828. <https://doi.org/10.1109/ACCESS.2024.3365368>

Price, M. (2024, May). *Crip Spacetime: Access, Failure, and Accountability in Academic Life*. Duke University Press. <https://doi-org.proxy.lib.sfu.ca/10.1215/9781478059370-001>

Rabinowitz, T., & Wallace, S. (2023, December 1). *New data on disability in Canada, 2022*. Government of Canada, Statistics Canada.

<https://www150.statcan.gc.ca/n1/pub/11-627-m/11-627-m2023063-eng.htm>

Rose, Geoffrey. A. (1993). *The strategy of preventive medicine* / Geoffrey Rose. Oxford University

Statistics Canada. (2023, May 31). Social connections among persons with and without mental health-related disabilities, 2020. Statistics Canada.

https://www150.statcan.gc.ca/n1/pub/89-654-x/89-654-x2023001-eng.htm?utm_source=chatgpt.com

Statistics Canada. (2023, December 1). Canadian Survey on Disability, 2017 to 2022. The Daily. <https://www150.statcan.gc.ca/n1/daily-quotidien/231201/dq231201b-eng.htm>

Statistics Canada. (2024, May 28). A demographic, employment and income profile of persons with disabilities aged 15 years and over in Canada, 2022. Statistics Canada.

https://www150.statcan.gc.ca/n1/pub/89-654-x/89-654-x2024001-eng.htm?utm_source=chatgpt.com

Tielman, M. L., Suárez-Figueroa, M. C., Jönsson, A., Neerinx, M. A., & Cavalcante Siebert, L. (2024). Explainable AI for all—A roadmap for inclusive XAI for people with cognitive disabilities. *Technology in Society*, 79, Article 102685.

<https://doi.org/10.1016/j.techsoc.2024.102685>

Treasury Board of Canada Secretariat. (2024, October 4). The Digital Nations. Government of Canada.

<https://www.canada.ca/en/government/system/digital-government/digital-nations.html>

Urbina, J. T., Vu, P. D., & Nguyen, M. V. (2025). Disability ethics and education in the age of artificial intelligence: Identifying ability bias in ChatGPT and Gemini. *Archives of Physical Medicine and Rehabilitation*, 106(1), 14-19. <https://doi.org/10.1016/j.apmr.2024.08.014>

Valdivia, A. (2024). The supply chain capitalism of AI: A call to (re)think algorithmic harms and resistance through environmental lens. *Information, Communication & Society*. Advance online publication.

<https://www.tandfonline.com/doi/full/10.1080/1369118X.2024.2420021#d1e235>